

Stress Classification Using a Low-Cost Optical Fiber Physiological Sensor: A Preliminary Study

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Abstract—The classification and measurement of stress through physiological sensors have been studied because of their high impact on mental health and well-being. Low-cost optical fiber wearable sensors, can monitor signals and be adaptive systems. Therefore, the present study sought to evaluate a low-cost optical fiber physiological sensor developed for stress classification while monitoring physiological signals. Five volunteers participated in a stress-induced test, where pulse and respiratory measurements were correctly performed with the optical fiber sensor during both stress and relaxation. The data obtained were used to find the best machine-learning classification algorithm for stress detection: a Bagged Decision Tree Classifier with an accuracy of 88%, and a precision of 93%. Thus, the proposed sensor can effectively monitor pulse and respiratory signals for stress classification. Based on these results, a study with more volunteers will be performed.

Index Terms—Optical fiber sensor, Physiological signals, Stress classification.

I. INTRODUCTION

The technology for measuring physiological variables is broad, and sensors are developed considering needs, applications, comfort, and cost. There are electrical [1], piezoresistive [2], and optical fiber sensors [3] that provide different benefits for the application and final objective of each sensor. When measuring and monitoring physiological signals, the main signs are derived from the cardiac activity and respiration [4].

Among the variables obtained from the cardiac activity is heart rate variability (HRV), which is associated with variations in the sympathetic system that reacts to stress and controls vital body functions [5]. Mental health has been a topic of recent interest due to the relationship between mental health, diseases, and quality of life, including the recent increase in the prevalence of anxiety disorders and depression directly related to stress factors [6]. Considering this, studies have been conducted to analyze, measure, classify, and detect stress using physiological sensors [7].

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The ECG sensors are highly effective in detecting the whole heart's electrical activity for determining the HRV features [1], but are costly, for clinical use, and use electrodes of one use on the participant's chests. In addition, other commercial sensors such as the Polar, the Garmin, and the Apple watch have been used for physiological measurement during sports. These sensors are highly effective in measuring the heart rate and have been tested as a tool for stress detection [5], giving good results. Moreover, some of them only give the heart variables, and not the signal for monitoring the sensor's behavior.

Among the optical fiber sensors developed for measuring these parameters can be highlighted the polymeric optical fiber (POF) [8] and Fiber Bragg Grating (FBG) sensors [10], [9]. FBGs are highly sensitive sensors, with electromagnetic immunity based on the variation in a specific wavelength linked to the grating inscribed on the optical fiber core. However, the interrogation methods for FBG sensors are costly and big-sized, which are in improvement for wearable applications [11]. Likewise, POF sensors use the variation in light intensity generated by sensitive areas that cause light losses which vary according to the pressure, expansion, movement, or angle of the sensitive areas [8], [3]. POF sensors are made of simple optical and electronic devices, which makes them low-cost sensors with simpler variation detection able to detect the RR peaks for the HRV analysis. POF sensors can be embedded in shirts [8] or mattresses [3], which are useful for various real-life applications, stress measurement and sleep monitoring.

Therefore, this project has the objective of developing a low-cost sensor capable of measuring respiration, pulse, and the variables necessary for stress classification, such as HRV, in order to implement the sensor in future work for continuous real-time monitoring of the presence of stress.

II. METHODOLOGY

A. The proposed optical fiber sensor

The proposed physiological optical fiber sensor consisted of a Poly(methyl methacrylate) (PMMA) POF (SH4001, Mitsubishi Chemical Co.), polymeric elastic transparent filament (Flex, 3D Fila), a photodetector (IF D93, Industrial Fiber Optics, Inc.), and a light-emitting diode (LED, IF E97, Industrial Fiber Optics, Inc.) which is in the red visible light at 650nm. The sensor works with the POF joined to the elastic filament with UV glue and a 3D printed piece design for the diameter of both elements (2.4 mm for the POF, and 1.75 mm for the filament). Two sensitive zones were created using this model, and a silicone coat was made in the sensitive

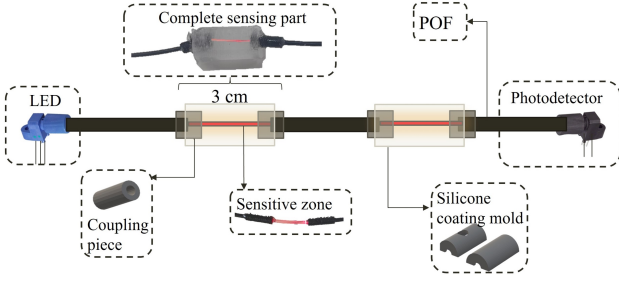


Fig. 1. Proposed optical fiber sensor operation schematic with the main components for physiological monitoring during the stress-induced test. zones to reduce light loss and allow light transmission through the filament. The sensitive zone works as the sensing part, the elastic filament with the silicone coat generates variations as the chest expands and contracts due to the heartbeat and respiration. The electronic circuit was composed of the LED, the photodetector, an amplification circuit optimized for the best signal detection of the sensor, and a Digital Acquisition System with 16-bits of resolution (DAQ, USB-6211, National Instruments) to send the digitalized signal to the computer and processing software. The sensor schematic is shown in Figure 1, where the light is coupled to the optical fiber, and the sensitive areas generate light loss depending on the pulse and breathing chest's movement in different frequencies, which is measured by the photodetector. The sensor was located over clothes, making it unobtrusive, and wearable, and was adapted to all volunteers with a Velcro flexible band. For the evaluation, the sensor was fitted to each participant's chest at axillary height and tightened according to the participant's comfort.

B. Participants

Five volunteers participated in the study (one female and four males, 22.6 ± 0.55 years). Each volunteer signed an informed consent form and the procedure was explained. A one-minute test was performed to obtain the signals and check the signal quality before the stress test. If the signal was abnormal, the sensor was adjusted, and the recording was repeated until the acquired signal was acceptable for analysis.

C. Stress test procedure

This work is the phase "Validation of the low-cost fiber optic physiological sensor" of a bigger protocol approved by the ethics committee of the Special School Club Leones Cruz del Sur. Initially, the data acquisition for the relaxation state was for five minutes. In this part of the test, the volunteer was seated in a comfortable chair and relaxing music was displayed. Subsequently, stress induction was performed with five one-minute tests. The word color test [10] of one minute twice, where the word of one color is displayed at 1.12 Hz in another color (e.g., the term "green" is displayed in orange, and the correct answer is "orange"). After that, the one-minute algebraic test was performed three times [10] in which participants were asked to mentally solve algebraic operations of low and mid-level (e.g., how much is 50×13 ? When the participant answers, the next algebraic operation is displayed, at the end of each minute the correct and incorrect answers

are shown). Finally, participants were asked to stay until initial heart and respiratory rates were recovered.

D. Data processing, features extraction, and machine learning (ML) algorithms

For the raw signal acquired from the sensor, two digital filters were applied to obtain pulse and respiratory signals. First, the pulse filter was a Butterworth third-order bandpass filter (1-3.5 Hz) [8]; and for the respiratory signal, a Butterworth third-order bandpass filter (0.1-0.8 Hz) was applied [8]. In addition, to obtain the heart and respiratory rates, the Findpeaks function of MATLAB was used with a minimum peak distance of 6 to avoid consecutive incorrect peaks from noise still present in the filtered signal and filter influence.

Twenty-two reference features [1], were extracted from the pulse and respiratory signals obtained. The main features extracted were the mean, standard deviation, median, skewness, kurtosis, and rms; and from the pulse signal, the inter-peak interval, and the heart rate variability (HRV). Subsequently, the samples were labeled as stress and non-stress data. Finally, 35 one-minute samples were used as training, and 15 samples were used for testing as a preliminary test of stress classification with the sensor. Classification algorithms were obtained using the MATLAB Classification Learner tool.

The main ML algorithms for stress detection reported were evaluated with the obtained set of data: decision trees, Gaussian and kernel Naive Bayes models, linear support vector machines, and KNN algorithms [1]. The metrics used for the evaluation of the algorithms by applying the training algorithm to the test data are accuracy, precision, recall, and F-score.

III. RESULTS AND DISCUSSION

A. Heart and respiratory rate detection

To validate the pulse and respiration measurement of the developed sensor, it was performed for one minute with each participant before the stress induction test. An example of ten seconds of the signals acquired with the sensor is shown in Figure 2. Compared to different commercial physiological detection devices [4], the sensor provides the graph of the physiological signals while also measuring both signals simultaneously. In some commercial sensors, only the heart rate over time is shown, which does not give the variation of peak-to-peak interval. Likewise, it is necessary to improve the sensitivity of the sensor and the preprocessing step to generate a clearer pulse and breathing wave for the identification of the peaks and valleys and to facilitate obtaining the features.

B. Stress classification

The best result for the stress classification validation was with a Bagged Decision Tree Classifier (BDTC) algorithm using 70% of data and likewise calculating the evaluation metrics with the test data. The results in the classification of the validation data were an accuracy of 88.4%, a precision of 93.8%, a recall of 83.3%, and an F-score of 88.2%. Within the comparison shown in Table I, other sensors and specific techniques used with public datasets reported similar results

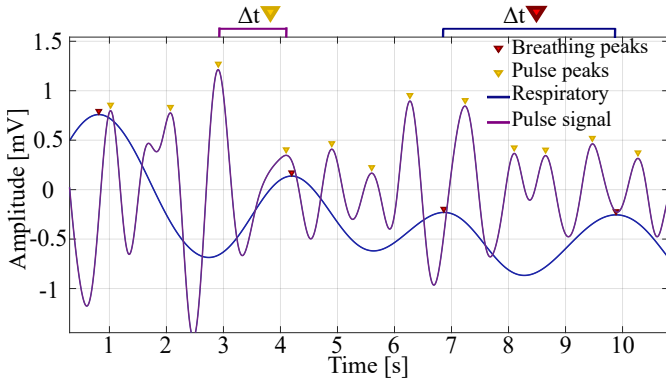


Fig. 2. Example of pulse and respiratory filtered signals obtained with the sensor, showing the peaks detection with the findpeaks Matlab function.

TABLE I
RESULTS COMPARISON WITH OTHER RELATED STUDIES INCLUDING ML ALGORITHMS FOR STRESS CLASSIFICATION

Volunteers	Sensor used	Algorithm	Accuracy	Study
College students	Developed POF low-cost sensor	BDTC	88.4%	This work
Police officers	Polar H10 HR monitor with a Polar Pro Chest Strap	SVM	86.5%	[5]
Police officers	Polar H10 HR monitor with a Polar Pro Chest Strap	SVM	90.5%	[5]
Spider-fearful subjects	BITalino biosignal measurement device	BDTC	89.9%	[1]

and accuracies. *Lee et al.* [5] applied traditional short HRV features and a novel ultrashort HRV feature for the estimation of police officers' stress with an accuracy of 86.5%, and 90.5%, respectively. Furthermore, *Ihmig et al.* [1], obtained an accuracy of 89.9% with their public database of spider-fearful subjects, and also obtained the best results with a BDTC.

Compared with what is reported with fiber optic sensors for stress detection, a detection analysis was found with the pulse through heart rate variability (HRV) features with the Poincaré plot analysis with two participants [10], where the influence of stress on HRV is graphically differentiated. However, no classification was performed with ML techniques, and differences were found between the two participants. There have been studies where the importance of separating subjects in extracting and analyzing features has been stated [7], which can be considered for an inter-subject analysis study with the proposed optical fiber sensor.

Considering this, Volunteer 4 had the best performance in the word color test, and Volunteer 3 had the best performance in the algebraic test. The performance results show the difference between volunteers in their commitment to the task and the difficulties in specific tasks, like Volunteer 5, who was motivated in the tasks but found the algebraic test more difficult than the word color test, whereby the performance in both tests was varied. In some other cases, there was no motivation to execute quickly the test, as in the

case of Volunteer 2. Even when the instructions were clearly explained, there were no real motivators to affirm commitment and effort in the tests, thereby, the stress induction.

Finally, the volunteers are college students, and the induction methods were not directly related to their real-life stressors, compared with other datasets where the induction tests were directly related to specific fears [1], guaranteeing the stress induction and obtaining better results in the classification. However, stress can be induced in other less-guaranteed cases, like frustration, social stressors, or real-life novel situations.

IV. CONCLUSIONS AND FUTURE WORK

With the data obtained from the proposed fiber optic sensor, a total of 50 one-minute samples were obtained, which is considerably less than the data in public databases [1]. Even though it is a small database, it can be segmented into smaller time windows (10, 20, or 30 seconds), which can provide more information and better classification results. The sensor was able to estimate stress 88% correctly with stress induction methods without much complexity using a BDTC. To improve this result, more complex stress induction tests can be implemented on a larger number of volunteers to obtain a more robust database that can more reliably estimate the presence or absence of stress. Moreover, a clear pulse and respiratory signal were obtained during the stress test, where the volunteers were talking when answering. Therefore, the proposed sensor can be used in future works as a wearable system for stress detection during real-life situations and in real-time scenarios.

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